NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2023-2024

EE2005 – ELECTRICAL DEVICES AND MACHINES

April / May 2024 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 4 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

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1. (a) The magnetic circuit shown in Figure 1 has a N-turn coil wound on the center limb. The circuit has an air gap of mean length $l_g = 1.25$ mm and the mean lengths of the left, center and right limbs are $l_1 = 75$ cm, $l_c = 30$ cm, and $l_2 = 65$ cm, respectively. The magnetic circuit has a uniform cross-sectional area of 25 cm^2.

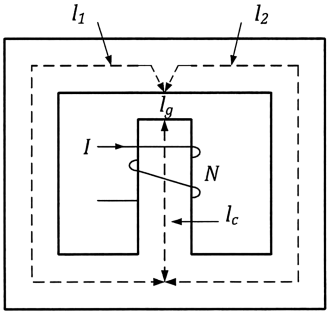


Figure 1

The B-H curve of the magnetic material used is given as follows:

B (T)	0.3	0.5	0.6	0.8	1.0	1.1
H (At/m)	500	650	750	900	1125	1350

Note: Question No. 1 continues page 2.

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(i) Draw the magnetic equivalent circuit of Figure 1. Given that the required flux density in the left limb is $B_1 = 0.5$ T, determine the flux densities in the center limb, the air gap, and the right limb. The permeability of free space μ_o is $4\pi x10^{-7}$ H/m. Ignore any leakage and fringing.

(ii) What is the current I in the coil needed to obtain the required flux density $B_1 = 0.5$ T in the left limb for $N = 500$?

(iii) What is the energy stored in the air gap?

(16 Marks)

(b) Figure 2 shows an electromagnet with a N-turn coil wound on the center limb of the immovable part. The air gap length (l_g m) and the mean lengths of the sections on the movable part are indicated ($l_m = 20$ cm). The magnetic circuit has a uniform cross-sectional area of 4 cm^2. The relative permeabilities μ_r of the immovable and movable parts are infinite and 1500, respectively.

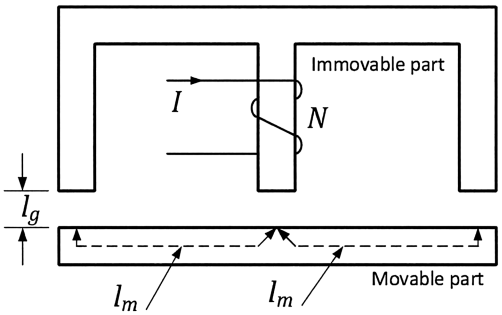


Figure 2

Determine the current I in the coil and the magnetic force F_m exerted on the movable part to keep it suspended in air at l_g= 0.01 m from the immovable part if $N = 120$ and the energy stored in the magnetic field is 1.5 mJ. Ignore any leakage and fringing. (Hint: First, draw the magnetic equivalent circuit and then find the total reluctance.)

(9 Marks)

2. Consider a 60 kVA, 2400/240-V single-phase transformer. Ignoring the no-load current, the short-circuit test was conducted with measurements taken on the high voltage side. The results are as follows:

$$V_{sc} = 75\,V, \quad I_{sc} = 25\,A, \quad P_{sc} = 1250\,W$$

(a) Draw the equivalent circuit of the transformer referred to the high voltage side, indicating the series impedance clearly.

(4 Marks)

Note: Question No. 2 continues page 3.

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(b) Three such single-phase transformers are used to form a Δ-Y connected transformer bank.

(i) Draw the per-phase Y-Y equivalent circuit referred to the high voltage side, indicating the series impedance clearly.

(ii) Using the equivalent circuit obtained in part (i), calculate the input line voltage and input power factor of the transformer bank when it delivers full load at 0.85 lagging power factor and rated secondary voltage. You may take the rated phase secondary voltage as reference, i.e., $\vec{V}_{\phi 2} = V_{\phi 2}\angle 0^o$.

(iii) Calculate the voltage regulation of the transformer bank.

(15 Marks)

(c) The single-phase transformer is connected as a 2400/2640 step up autotransformer. Draw the diagram of the autotransformer and determine its kVA rating and the output at full load, 0.85 power factor lagging.

(6 Marks)

3. (a) A three-phase, 460-V, 50-Hz, 4-pole, wye-connected induction motor has the following per phase parameters referred to the stator:

$$R_1 = 0.5\ \Omega,\ X_M = 50\ \Omega,\ X_1 = 1\ \Omega,\ X_2 = 1\ \Omega,\ R_2 = 0.28\ \Omega$$

The rotational losses are 1200 W. Determine

(i) The magnitude of the input current at starting.

(ii) The slip, output power and efficiency of the motor when it is running at the rated speed of 1458 rpm.

(12 Marks)

(b) The following tests were conducted on a three-phase, 460-V, 6-pole, 50-Hz, wye-connected induction motor:

Locked rotor test: 100 V, 140 A, 8200 W

DC test: 22 V, 138 A

Determine the per-phase parameters R_1, X_1, X_2 and R_2 of the induction motor equivalent circuit referred to the stator. Assume that $X_1 = X_2$.

(6 Marks)

Note: Question No. 3 continues page 4.

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(c) Consider a three-phase wye-connected induction motor which has negligible stator impedance, i.e. $R_1 = 0\ \Omega$ and $X_1 = 0\ \Omega$. If the maximum torque developed by the motor is 150 % of the starting torque developed, determine the slip at which the torque developed is maximum.

(7 Marks)

4. (a) A dc shunt motor has an armature winding resistance and a field winding resistance of $0.3\ \Omega$ and $240\ \Omega$, respectively. The armature terminals are connected to a 240-V dc supply.

(i) At full load, the motor is rotating at a speed of 1180 rpm and draws an armature current of 38 A. Determine the torque developed by the motor at full load.

(ii) If the load is now disconnected from the motor shaft and the motor draws an armature current of 4 A, determine the rotational losses and no-load speed of the motor.

(iii) An external variable resistance R_{ext} is now connected in series with the field winding. If the power developed by the motor is required to be 8000 W when rotating at a speed of 1280 rpm, determine the value of R_{ext} required. Assume that the motor is operating in the linear part of the magnetization curve.

(15 Marks)

(b) A separately-excited dc generator has an armature winding resistance and a field winding resistance of $0.3\ \Omega$ and $150\ \Omega$, respectively. The machine is driven by an external prime mover at a constant speed and the effect of armature reaction can be neglected. The constant rotational losses of the machine are 300 W. A load is connected to the armature terminals and an external variable resistance R_{ext} is connected in series with the field winding to a 200-V dc supply.

(i) If R_{ext} is adjusted such that the field current is 0.7 A and the internal generated voltage is 150 V, determine the efficiency of the generator when it is delivering an armature current of 14.5 A.

(ii) If the generator is delivering an output power of 3150 W at a terminal voltage of 175 V, determine the value of R_{ext} required. Assume that the machine is operating in the linear part of the magnetization curve.

(10 Marks)

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.

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